

Dostarlimab in Human Breast Cancer

Drug Summary

Dostarlimab (Jemperli) is a humanized IgG4 monoclonal antibody targeting the PD-1 immune checkpoint receptor ¹. It is approved for mismatch repair-deficient (dMMR) endometrial cancer and other dMMR solid tumors ¹ ². While not yet specifically approved in breast cancer, dostarlimab and other PD-1 inhibitors are under active investigation across breast cancer subtypes. Mechanistically, by binding PD-1 on T cells and blocking its interaction with PD-L1/PD-L2, dostarlimab releases the “brakes” on cytotoxic T-lymphocytes, restoring anti-tumor immune activity ³. In breast oncology, immune checkpoint blockade has shown the most benefit in highly immunogenic subtypes like triple-negative breast cancer (TNBC) and in tumors expressing PD-L1 ⁴ ⁵. For example, adding the PD-1 inhibitor pembrolizumab to chemotherapy significantly improves outcomes in PD-L1-positive metastatic TNBC ⁴ and is now part of the standard of care. TNBC and HER2-positive tumors tend to have higher tumor-infiltrating lymphocytes and immune gene signatures, correlating with better responsiveness to PD-1 blockade ⁵. In contrast, hormone receptor-positive/HER2-negative (HR+/HER2-) breast cancers are typically less immunogenic, and single-agent PD-1 inhibitors have shown limited activity ⁵. Nevertheless, combinations are being explored: for instance, in a Phase II trial for high-risk early HR+/HER2- breast cancer, adding a PD-1 inhibitor to neoadjuvant chemotherapy significantly increased pathological complete response rates ⁶. Similarly, ongoing trials in HER2-positive disease (e.g. combining PD-1 blockade with anti-HER2 therapy) are examining whether immune checkpoint inhibition can enhance responses in HER2-driven tumors. Overall, dostarlimab represents a promising immunotherapeutic approach in breast cancer, especially in subsets with pre-existing anti-tumor immunity (e.g. PD-L1-expressing or **BRCA1/2**-mutated tumors) ⁷ ⁸. Its role in treatment algorithms is emerging through clinical trials and will likely mirror class effects seen with other PD-1 inhibitors, with greatest impact in TNBC and selected biomarker-enriched populations.

Identifiers & Synonyms

- **ChEMBL ID:** ChEMBL4298124 ⁹
- **DrugBank ID:** DB15627 ⁹
- **Synonyms/Trade Names:** Dostarlimab-gxly (USAN); **Jemperli®** (brand name) ¹⁰ ¹¹; TSR-042 ¹⁰ ¹¹; GSK-4057190 (GSK's development code) ¹²; ANB-011 (AnaptysBio code) ¹³; WBP-285 ¹⁰.
Dostarlimab-gxly is the full nonproprietary name used in regulatory labels ¹⁴.

Mechanism of Action (Breast Cancer Context)

Dostarlimab's mechanism of action is to inhibit PD-1, thereby unleashing T-cell mediated immune responses against tumor cells. PD-1 (gene **PDCD1**) is an inhibitory receptor on activated T cells; when engaged by its ligands PD-L1/PD-L2 on tumor or immune cells, it attenuates T-cell proliferation and cytokine production ³. Dostarlimab binds to PD-1 and prevents this interaction, which can restore T-cell activation and tumor cell killing ³. In breast cancer, the efficacy of PD-1 blockade is closely tied to the tumor immune microenvironment. TNBC and HER2-enriched subtypes often have “hot” tumors with high levels of tumor-infiltrating lymphocytes (TILs), PD-L1 expression, and interferon-gamma-stimulated genes ⁵. In these

contexts, PD-1 inhibitors (like dostarlimab) can produce significant anti-tumor activity by reactivating endogenous T cells. Clinical trials in metastatic TNBC have shown that adding PD-1/L1 antibodies to chemotherapy improves survival **predominantly in PD-L1-positive tumors**, reflecting the requirement for an existing anti-tumor immune response ⁴. By contrast, HR+/HER2- (luminal A/B) breast cancers tend to be immunologically “cold” with low TILs and PD-L1, and single-agent PD-1 blockade yields modest benefit ⁵. Combination strategies aim to overcome this: for example, PARP inhibitors can induce cytosolic DNA and activate the cGAS–STING pathway, promoting T-cell recruitment, which provides a strong rationale for combining PARP inhibition with PD-1 blockade in BRCA-mutated or homologous recombination-deficient breast cancers ⁸. Early trials of niraparib (PARP inhibitor) plus dostarlimab in **gBRCA1/2-mutated** breast cancer have shown increased TILs and some pathological responses even in ER+ tumors ^{7 15}, supporting this mechanistic synergy. Likewise, in HER2-positive breast cancer, PD-1 inhibitors are being combined with HER2-targeted therapies to augment antibody-dependent cellular cytotoxicity and T-cell responses (e.g. pembrolizumab + trastuzumab in trials) ^{16 17}. Overall, dostarlimab’s anti-tumor effect in breast cancer arises from reinvigorating T cells within the tumor microenvironment, and its success depends on tumor- and host-specific factors such as PD-L1 status, TIL density, antigen presentation competence, and the presence of immunosuppressive pathways.

Primary Targets (Human)

Dostarlimab’s direct molecular target is **PD-1** (programmed cell death protein 1, gene **PDCD1**, also known as CD279). PD-1 is an immune checkpoint receptor expressed on T cells (as well as B cells and NK cells) that downregulates immune responses upon ligand binding ³. By targeting PD-1, dostarlimab blocks its interaction with PD-L1 (CD274) and PD-L2, thereby preventing the inhibitory signaling cascade that normally dampens T-cell activity ³. The result is enhanced T-cell activation and cytokine release, enabling T cells to attack cancer cells. **PDCD1** is the only primary target of dostarlimab; it does not bind PD-L1 or other receptors. This specificity is supported by biochemical and clinical data from the GARNET trial and others showing that dostarlimab’s therapeutic effects (and immune-related side effects) are consistent with PD-1 blockade ¹. In summary, **PD-1 (PDCD1)** is the principal target of dostarlimab in humans, and its inhibition underlies the drug’s mechanism of action in breast cancer and other tumors.

Pathways (High-Level Overview)

Dostarlimab’s immunotherapeutic action in breast cancer engages a network of immune and tumor cell pathways:

- **PD-1 Checkpoint Signaling (Reactome: *CO-INHIBITION_BY_PD_1*)** – *Immune checkpoint pathway*. Dostarlimab blocks the PD-1/PD-L1 inhibitory axis, thereby **upregulating T-cell receptor (TCR) signaling** and effector T-cell activity ³. This pathway is central to its MoA: by preventing PD-1 mediated co-inhibition, dostarlimab promotes T-cell proliferation, IL-2 production, and cytotoxic functions that can lead to tumor cell death ³.
- **T-cell Activation Pathways (KEGG: *hsa04660 T cell receptor signaling*)** – *Immune activation*. Effective PD-1 blockade requires an active T-cell receptor pathway. In responders, **antigen presentation and TCR engagement** are intact, leading to robust T-cell activation when PD-1 is uninhibited ¹⁸. Conversely, defects in antigen presentation (e.g. loss of MHC class I) can make tumors resistant by *downregulating* this pathway ¹⁹.

- **Interferon Gamma Response (MSigDB Hallmark: *HALLMARK_INTERFERON_GAMMA_RESPONSE*)** – *Pro-inflammatory/Th1 immunity*. This pathway is typically **upregulated in tumors that respond to PD-1 blockade**, reflecting a pre-existing immune-inflamed microenvironment. Markers of an IFN- γ response (e.g. CXCL9/10, PD-L1) correlate with sensitivity to dostarlimab and other ICIs ⁴. A high “T-cell-inflamed” gene expression profile (rich in IFN- γ -stimulated genes) is a positive predictive biomarker for PD-1 inhibitor efficacy ⁴.
- **cGAS-STING Pathway (KEGG: *hsa04623 Cytosolic DNA-sensing pathway*)** – *Innate immune sensing*. Upregulation of the STING (stimulator of interferon genes) pathway in tumor cells or dendritic cells can increase immunogenicity. DNA damage (e.g. from chemotherapy or BRCA1/2-related defects) activates cGAS-STING, leading to type I interferon production and T-cell recruitment. Activation of this pathway **enhances sensitivity** to PD-1 blockade ⁸. Conversely, tumors with epigenetic loss of STING signaling have poor TIL infiltration and resistance ²⁰.
- **PI3K/AKT/mTOR Pathway (MSigDB Hallmark: *HALLMARK_PI3K_AKT_MTOR_SIGNALING*)** – *Growth/survival signaling*. Hyperactivation of the PI3K-AKT pathway (often via PTEN loss) in tumors is associated with an **immunosuppressive microenvironment** and resistance to PD-1 therapy ²¹. PI3K/AKT signaling in tumor cells can lead to lower antigen presentation and recruitment of suppressive immune cells, thereby *blunting* T-cell-mediated killing ²¹. In contrast, inhibition or downregulation of PI3K-AKT can improve T-cell infiltration and immunotherapy response ²¹.
- **WNT/ β -Catenin Signaling (KEGG: *hsa04310 Wnt signaling pathway*)** – *Developmental/immune evasion pathway*. Upregulation of Wnt/ β -catenin signaling in breast tumors (e.g. basal-like cancers) fosters immune evasion by excluding dendritic cells and T cells from the tumor microenvironment ²¹. A *high Wnt* pathway activity is linked to “cold” tumors and resistance to checkpoint inhibitors. Conversely, *downregulating* Wnt/ β -catenin signaling can help convert tumors to a more immune-permissive state, potentially increasing sensitivity to dostarlimab.
- **TGF- β Signaling (MSigDB Hallmark: *HALLMARK_TGF_BETA_SIGNALING*)** – *Immunosuppressive cytokine pathway*. TGF- β is often **upregulated in resistant tumors**, promoting fibrosis and immune exclusion. In breast cancer, TGF- β signaling induces factors like indoleamine 2,3-dioxygenase (IDO1) and the chemokine CCL22, which recruit regulatory T cells and suppress effector T cells ²². High TGF- β activity can thus counteract PD-1 blockade (a phenomenon noted in immune-excluded tumors). Combination approaches targeting TGF- β alongside PD-1 (or using TGF- β blocking bifunctional antibodies) are being explored to overcome this resistance mechanism.
- **IL-6/JAK-STAT3 Pathway (ImmunoSigDB: *SIGNALING_BY_IL6*)** – *Inflammation and immune regulation*. Chronic IL-6/STAT3 activation in the tumor microenvironment supports expansion of myeloid-derived suppressor cells (MDSCs) and M2 macrophages, leading to T-cell suppression. STAT3 and related pathways are **upregulated in some immune-resistant breast cancers**, contributing to a protumoral, immunosuppressive niche ²¹. Inhibiting IL-6 or STAT3 signaling can relieve some of this immunosuppression, thereby potentially increasing responsiveness to PD-1 blockade.
- **Antigen Presentation & Processing (GO: *0019882 Antigen presentation via MHC I*)** – *Immune recognition*. Proper antigen processing and presentation on MHC class I is crucial for T-cell mediated tumor killing. Tumors that lose MHC class I expression or downregulate antigen presentation genes (through mutations in B2M, TAP1, etc.) can escape immune surveillance ¹⁹. Such *downregulation* of

the antigen presentation pathway is a known mechanism of resistance to ICIs. Conversely, breast tumors with intact or upregulated antigen presentation machinery (e.g. high MHC I, immunoproteasome activity) provide the necessary targets for cytotoxic T cells unleashed by dostarlimab, and thus respond better.

- **Apoptosis and Cell Cycle Pathways (MSigDB: *HALLMARK_APOPTOSIS*, *HALLMARK_E2F_TARGETS*)** – *Downstream tumor responses.* Although not direct targets of dostarlimab, these pathways indicate how tumors respond once immune pressure is applied. Successful PD-1 blockade can lead to increased tumor cell apoptosis (via granzymes, Fas/FasL interactions) and cell cycle arrest as immune-mediated damage accumulates. Upregulation of pro-apoptotic signaling in tumors (e.g. via IFN- γ and TNF α from T cells) is associated with sensitivity (tumor regression), whereas tumors that quickly upregulate cell-cycle and survival signals (via MYC, E2F, etc.) may resist immune-mediated killing. These downstream pathways illustrate the interplay between immune attack and tumor-intrinsic survival programs under checkpoint therapy.

(The above are key pathways perturbed by dostarlimab's action in breast cancer. Some are mechanistically linked to PD-1 (e.g. T-cell activation, IFN signaling), while others modulate sensitivity (e.g. cGAS-STING upregulation enhances response) or resistance (e.g. WNT, TGF- β create immune evasion). Pathway relevance may vary by subtype: for instance, BRCA-mutant and HRD+ tumors rely on cGAS-STING and DNA repair defects to attract T cells, HER2+ tumors often involve PI3K/AKT cross-talk with immunity, and luminal tumors may have prominent STAT3/TGF- β immunosuppressive signaling.)

Upregulated Pathways (Response to Therapy)

Certain pathways are found to be transcriptionally **upregulated in the context of PD-1 blockade** in breast cancer, either as part of the anti-tumor response or as compensatory resistance mechanisms:

- **Inflammatory/Interferon Pathways** – In responders, PD-1 blockade drives upregulation of interferon-inducible genes and cytotoxic immune pathways. For example, treatment can elevate **IFN- γ -responsive gene sets** (CXCL9, CXCL10, MHC molecules, etc.) in the tumor microenvironment, reflecting T-cell activation ⁴. In a neoadjuvant trial, patients responding to niraparib + dostarlimab showed increased TILs and immune gene signatures after 3 weeks of therapy ¹⁵, indicating an *upregulation* of immune pathways (such as T-cell receptor signaling and Th1 cytokines) early on.
- **T-cell Checkpoint and Activation Markers** – Paradoxically, PD-1 blockade can induce *upregulation of other inhibitory checkpoints* as a feedback mechanism. An example is **TIM-3 (HAVCR2)** on T cells: in the setting of PD-1 therapy, tumor-infiltrating lymphocytes may increase TIM-3 expression via PI3K/Akt pathway activation ²³. This TIM-3 upregulation has been observed in other cancers (e.g. head/neck) as an adaptive resistance, and similar findings are reported in breast cancer models (e.g. increased CTLA-4, LAG-3 on T cells after PD-1/PD-L1 blockade) ²⁴. Thus, while cytotoxic T cells are activated (upregulating *effector* pathways), they may concurrently upregulate alternate checkpoints, a sign of incipient T-cell exhaustion.
- **cGAS-STING and Type I IFN** – In tumors with prior DNA damage or ongoing response, PD-1 blockade can amplify innate immune sensing. Upregulation of **STING pathway genes** (STING1, IFNB1) has been noted in responders, particularly in HRD-positive breast cancers. The combination of PARP inhibition and dostarlimab in BRCA-mutated patients, for instance, showed evidence of

cGAS–STING activation (elevated interferon signaling, increased TILs) after therapy ⁸. This suggests that STING-mediated **inflammasome** and dendritic cell activation pathways are upregulated during effective treatment, promoting an immune inflamed microenvironment.

- **Cytotoxic Effector Pathways** – Successful immune attack is marked by upregulation of **apoptosis and cytotoxicity pathways** in tumors. Transcriptional analyses show that with PD-1 blockade, breast tumors (especially TNBC) that respond have higher expression of granzyme B, perforin, Fas ligand, and other apoptosis mediators. These reflect the *upregulation* of the **granule-mediated killing pathway** in CD8⁺ T cells and NK cells. Concurrently, tumor cells in responders may show upregulation of pro-apoptotic signals (e.g. BAX, TRAIL pathways) as they undergo immune-mediated cell death (as reported in gene signatures of pCR tissues).
- **MHC II and Antigen Presentation** – There is evidence that PD-1 inhibitor therapy can increase expression of MHC class II on tumor cells or antigen-presenting cells in the tumor. For instance, an increase in HLA-DR and other MHC-II genes was observed in some TNBC responders receiving neoadjuvant anti-PD-1 (likely due to IFN- γ in the microenvironment). This *upregulated antigen presentation* might further enhance T-cell priming in situ. (By contrast, non-responders often fail to upregulate antigen presentation, or even downregulate it, as a mode of escape.)

In summary, when dostarlimab is effective, we see upregulation of immune-effector pathways (TCR signaling, IFN γ response, cytotoxic granules) and sometimes concurrent induction of alternative checkpoints (TIM-3, LAG-3) as an adaptive change ²⁴ ²³. Understanding these upregulated signals helps identify pharmacodynamic markers of response and potential combination targets (e.g. TIM-3 inhibitors to complement PD-1 blockade).

Downregulated Pathways

Exposure to dostarlimab can also lead to **downregulation of certain pathways** in breast tumors or immune cells, which may influence sensitivity or resistance:

- **Cell Proliferation and DNA Repair** – On-treatment biopsies from responders often show tumors with suppressed proliferation pathways. For instance, gene sets like E2F targets, MYC targets, and G₂/M cell-cycle checkpoint genes are *downregulated* in tumors that responded to PD-1 therapy (consistent with immune-mediated growth arrest). This is evidenced by lower Ki-67 and reduced expression of replication genes in post-therapy specimens achieving pathological complete response (pCR) ⁷ ¹⁵ ²³. Similarly, if immune pressure is effective, the tumor may downregulate DNA damage repair pathways (because cells undergoing apoptosis or being eliminated do not engage repair) – an observation in some TNBC transcriptomic analyses after immunotherapy.
- **Immunosuppressive Factor Production** – Checkpoint blockade can cause a decrease in certain immunosuppressive pathways, especially if those pathways are dominated by tumor or stromal signaling that is no longer stimulated by tumor presence. For example, **TGF- β signaling can be functionally downregulated** if effective immune clearance removes sources of TGF- β (like tumor cells or cancer-associated fibroblasts). In immune-“hot” tumors responding to therapy, one might see *lower expression of TGF- β -driven gene signatures* and cytokines like IL-10, as the tumor microenvironment shifts to a Th1-dominant state. (Conversely, in non-responders, TGF- β remains high or even increases, indicating ongoing immunosuppression ²².)

- **Metabolic and Angiogenic Pathways** – Some metabolic pathways in tumors get downregulated as an indirect effect of immune attack. For instance, breast tumors responding to PD-1 blockade have shown reduced expression of **tryptophan-catabolic enzymes (IDO1)**, which is normally upregulated by tumor and stromal cells to suppress T cells. Effective therapy may alleviate the IFN-driven feedback loop that induces IDO, thus *downregulating the kynurenine pathway*. Additionally, pro-angiogenic signals (VEGF, etc.) can decrease when immune cytokines like IFN-γ and TNF start normalizing the vasculature. This *downregulation of angiogenesis-related genes* has been reported in some responders, potentially improving T-cell infiltration further.
- **Alternative Immune Checkpoints (in TME)** – In the context of combination therapy or over time, an effective PD-1 blockade might actually reduce the need for certain compensatory pathways. An interesting example: if TIM-3⁺ exhausted T cells are reinvigorated and expand, subsequent treatment targeting TIM-3 could lead to *downregulation of TIM-3 pathway* activity. While not directly caused by dostarlimab alone, combination treatments aim to achieve such concurrent downregulation of multiple checkpoints. (Monotherapy with dostarlimab typically *increases* alternate checkpoints as noted, but the eventual goal is to suppress those via additional agents.)
- **MHC Downregulation in Resistant Tumors** – A critical resistance mechanism is tumor downregulation of antigen presentation (e.g. loss of MHC-I). Indeed, analyses of resistant breast tumors (progressing on PD-1 therapy) find *decreased expression of HLA class I molecules and antigen-processing genes* ¹⁹. This is a **negative** adaptive change: tumors that progressed had effectively silenced MHC-I, escaping T-cell killing. It highlights that in cases of non-response or acquired resistance, key immune pathways are actively *downregulated by the tumor*. (In a responsive scenario, we would conversely want these pathways maintained or upregulated.)

Overall, beneficial responses to dostarlimab correlate with downregulation of tumor-intrinsic programs (proliferation, DNA repair) and some immunosuppressive mediators, whereas resistance can involve tumor-driven downregulation of immune-recognition pathways (like MHC). Monitoring these changes via biopsies and RNA profiling helps identify whether a tumor is mounting resistance (e.g. by losing antigen presentation) or succumbing to immune pressure (e.g. by shutting down cell-cycle progression).

Sensitivity Pathways (Mechanisms of Sensitivity)

Mechanisms that increase sensitivity to dostarlimab in breast cancer often involve the upregulation or maintenance of pro-immune pathways and the attenuation of immunosuppressive signals. Key sensitivity-associated pathways include:

Sensitivity Pathways – Upregulation

These are pathways whose *upregulation* (higher activity/expression) **enhances the tumor's responsiveness** to dostarlimab:

- **Type I/II Interferon Signaling:** An active interferon-rich microenvironment strongly predicts benefit from PD-1 blockade. High baseline expression of an **IFN-γ-related gene signature** (e.g. CXCL9, CXCL10, IDO1, MHC-II genes) is associated with greater sensitivity to pembrolizumab in breast cancer ⁴. Upregulation of IFN signaling leads to increased antigen presentation and T-cell recruitment. For example, TNBC tumors with a “T-cell-inflamed” gene profile (often due to IFN-γ

upregulation from existing TILs) have significantly improved response rates to anti-PD-1 therapy ⁴. *Biological rationale:* IFNs promote tumor visibility to the immune system (via MHC upregulation) and activate cytotoxic Th1 immunity, thereby priming the tumor for PD-1 blockade.

- **cGAS–STING Pathway Activation:** Upregulation of the STING pathway (triggered by cytosolic DNA) enhances sensitivity to dostarlimab. HRD+ and BRCA-mutant breast cancers often have spontaneous DNA breaks that activate cGAS–STING, resulting in a baseline influx of immune cells. Preclinical and clinical data show that **stimulating STING** (through DNA damage or STING agonists) leads to robust CD8⁺ T-cell priming and makes tumors more susceptible to PD-1 blockade ⁸. In a clinical example, BRCA2-mutated ER+ breast cancers (normally less immunogenic) responded to PARP inhibitor + dostarlimab, presumably via PARP-induced upregulation of the cGAS–STING–IFN pathway ⁸. *Rationale:* STING upregulation turns “cold” tumors “hot” by producing Type I IFNs and dendritic cell activation, thus setting the stage for effective PD-1 blockade.
- **Tumor Mutational Burden & Neoantigen Pathways:** High tumor mutational burden (TMB) is often coupled with upregulated neoantigen presentation pathways, which correlates with sensitivity to ICIs. In breast cancer, tumors with **DNA repair defects** (BRCA1/2, POLE mutations, etc.) accumulate mutations and neoantigens. This tends to upregulate pathways like the **DNA damage response** (DDR) and downstream immune sensing. Clinically, MSI-high (mismatch repair–deficient) or TMB-high breast cancers (though rare) have shown deep and durable responses to PD-1 inhibitors ² ²⁵. *Rationale:* A high neoantigen load upregulates anti-tumor immunity (T cells recognizing more targets), making PD-1 blockade more effective. Pathways like **HALLMARK_DNA_REPAIR** being deficient/upregulated in response can indirectly favor immune recognition.
- **Th1/Cytotoxic T-cell Pathways:** Pathways indicative of active cytotoxic T cells – such as **Granzyme/Perforin pathway**, **Fas–FasL apoptotic signaling**, and **TNFα/NF-κB (immune effector)** pathways – if upregulated, denote a microenvironment already attacking the tumor. Breast tumors with elevated baseline CD8⁺ T-cell signatures and cytotoxic effector genes respond better to dostarlimab. One metric is TILs: a high stromal TIL count (reflecting upregulated local immune response) is an established marker of PD-1 inhibitor sensitivity in TNBC ²⁶. In the I-SPY2 trial and others, **upregulated cytotoxic activity** (e.g. granzyme B⁺ cell density) pre-treatment correlated with higher pCR rates under checkpoint therapy. *Rationale:* These pathways indicate the immune system is already engaged – PD-1 blockade then acts as an accelerant to an ongoing anti-tumor immune attack.
- **Chemokine Pathways (CXCR3/CXCL9 axis):** Upregulation of T-cell recruiting chemokines such as CXCL9, CXCL10 (which attract CXCR3⁺ Th1 cells) and CCL5 is linked to better responses. Many of these chemokines are part of the “**Allograft Rejection**” and **Inflammatory Response** gene sets that are prognostic for ICI benefit. Breast tumors that naturally express high levels of CXCL9/CXCL10 have more T-cell infiltration and respond better to PD-1 blockade (this has been observed in TNBC gene analyses). *Rationale:* Chemokine upregulation ensures effector T cells can traffic into the tumor; dostarlimab then sustains their function to mediate tumor killing.

Sensitivity Pathways – Downregulation

These are pathways whose *downregulation* (lower activity) **increases sensitivity** to dostarlimab – typically pathways that, when active, inhibit effective immunity:

- **TGF- β Signaling Suppression:** Downregulating the TGF- β pathway can markedly improve sensitivity to PD-1 blockade. TGF- β is a key driver of immune exclusion; high TGF- β in tumors correlates with poor responses to atezolizumab in multiple studies (e.g. in TNBC with a TGF- β -dominant stromal signature). Conversely, **inhibition or low activity of TGF- β** (either via genetic context or therapeutic blockade) allows T cells to penetrate tumors and synergize with PD-1 inhibitors ²². Indeed, a trial combining a TGF- β trap with anti-PD-L1 in metastatic TNBC (IMpassion130 exploratory analysis) suggested patients with low TGF- β signatures derived greater benefit. *Rationale:* Downregulating TGF- β removes a major immunosuppressive brake (fewer Tregs, less exclusion of T cells), thereby enhancing the efficacy of dostarlimab.
- **β -Catenin/Wnt Pathway Downregulation:** Tumors with loss-of-function in β -catenin or otherwise low Wnt signaling are more sensitive to immunotherapy. Preclinical models demonstrate that **active Wnt/ β -catenin signaling excludes dendritic cells**, leading to lack of T-cell priming. In basal-like breast cancers, high Wnt pathway activity is associated with immune-cold tumors; thus, tumors with *downregulated Wnt signaling* (either inherently or via tankyrase inhibitors in trials) show increased TILs and responsiveness to ICIs ²¹. *Rationale:* Downregulating Wnt removes a mechanism of immune evasion, permitting better T-cell infiltration and priming, which aids PD-1 blockade activity.
- **PI3K/AKT Pathway Downregulation:** Loss of the PI3K-AKT oncogenic pathway (e.g. PTEN intact or PI3K α mutant tumors treated with PI3K inhibitors) can improve immunotherapy response. Clinically, PTEN-positive tumors have been noted to respond better to PD-1/PD-L1 inhibitors than PTEN-null tumors in some analyses (as seen in melanoma and suggested in TNBC). In breast cancer, combining a PI3K inhibitor (alpelisib or idelalisib) with anti-PD-1 is hypothesized to enhance efficacy by **downregulating tumor-intrinsic PI3K signals**, thereby reducing immunosuppressive cytokine output and MDSC recruitment ²⁷ ²¹. Early-phase trials (e.g. taselisib+pembrolizumab) have shown improved immune activation in patients. *Rationale:* Downregulation of PI3K-AKT signaling in tumor cells can lead to a more immune-friendly tumor milieu (less VEGF, less IL-10, more T-cell access) and prevents the emergence of immunotherapy resistance associated with PI3K activation.
- **IDO/Kynurenine Pathway Inhibition:** The enzyme IDO1 (indoleamine 2,3-dioxygenase) in the tumor microenvironment consumes tryptophan and produces immunosuppressive metabolites (kynurenines). High IDO1 activity correlates with immune suppression and can be induced by IFN- γ as a negative feedback. Thus, **downregulating IDO1 activity** (for instance, using an IDO inhibitor or in tumors with naturally low IDO expression) may improve responses to dostarlimab. Although a Phase III trial of an IDO inhibitor + pembrolizumab failed in melanoma, in breast cancer the concept remains that tumors not upregulating IDO (or patients with low baseline IDO gene expression) tend to do better with checkpoint blockade ²². *Rationale:* IDO1 depletes tryptophan and expands regulatory T cells; removing this pathway can enhance T-cell function. (Notably, in some TNBC studies, high baseline IDO1 was associated with worse outcomes on ICIs, aligning with this rationale.)

- **VEGF/Angiogenesis Pathway Suppression:** Reducing pro-angiogenic signaling can improve immune cell infiltration and thus sensitivity to PD-1 blockade. While direct clinical evidence in breast cancer is still emerging, parallels from other cancers suggest that **downregulation of VEGF/VEGFR pathway** (through agents like bevacizumab or tyrosine kinase inhibitors) normalizes tumor vessels and relieves hypoxia, making the microenvironment more permissive for T-cell entry. Trials combining anti-VEGF with PD-L1 blockade in breast cancer (e.g. atezolizumab+bevacizumab in TNBC) have indicated enhanced immune activation, although outcomes are mixed. *Rationale:* An “angiogenic switch-off” turns a immune-excluded tumor into one accessible to effector cells; thus antiangiogenic therapy can potentiate PD-1 inhibitors.

(In summary, breast tumors are most sensitive to dostarlimab when pro-immune pathways (IFN, STING, T-cell chemokines) are up, and immunosuppressive pathways (TGF- β , Wnt, PI3K, IDO, etc.) are kept down. This balance creates an environment where T cells can effectively attack the cancer upon PD-1 blockade. Recognizing these pathways guides combination therapies: e.g., adding STING agonists or IFN modulators to boost response, or using TGF- β or PI3K inhibitors to alleviate resistance.)

Resistance Pathways (Mechanisms of Resistance)

Breast tumors can evade or become refractory to dostarlimab through various **resistance mechanisms**, often involving pathway alterations that blunt immune attack. Key resistance-associated pathways include:

Resistance Pathways – Upregulation

Pathways whose *upregulation* can drive resistance to dostarlimab (by helping the tumor escape immune destruction):

- **Alternate Immune Checkpoints (TIM-3, LAG-3, CTLA-4):** As mentioned, an adaptive resistance to PD-1 blockade is the upregulation of other inhibitory receptors. **TIM-3** upregulation on T cells (and sometimes on tumor or myeloid cells) is a well-documented resistance pathway ²³. In breast cancer, post-therapy biopsies from non-responders have shown increased expression of TIM-3 and LAG-3 on TILs, indicating T-cell exhaustion despite PD-1 blockade. Similarly, the tumor can upregulate **PD-L1** itself or on other immune cells (though PD-L1 is the target, extremely high PD-L1 might reflect immune pressure but also local immune suppression until drug binding). *Rationale:* The tumor exploits redundant checkpoint pathways; once PD-1 is blocked, TIM-3 (or LAG-3, etc.) can inhibit T cells via alternative mechanisms. This has led to combination trials (e.g. PD-1 + LAG-3 inhibitors) in immune-resistant TNBC.
- **TGF- β Pathway Activation:** High **TGF- β signaling** is a known driver of resistance. In immune-excluded TNBCs, a TGF- β “signature” in fibroblasts correlates with lack of response to atezolizumab ²². TGF- β upregulation leads to dense stroma and exclusion of T cells from the tumor core. For instance, in breast tumors that did not respond to dostarlimab, analysis often finds elevated TGF- β 1 and downstream SMAD target genes. Moreover, TGF- β induces IDO and the chemokine CCL22 (recruiting Tregs), compounding immunosuppression ²⁸. *Rationale:* Upregulated TGF- β essentially “walls off” the tumor from immune attack and polarizes the TME toward healing and fibrosis rather than inflammation, thereby undermining PD-1 blockade.

- **Wnt/ β -Catenin Signaling: Wnt pathway activation** is another route to resistance. Tumors with aberrant β -catenin activation lack intratumoral dendritic cells, resulting in poor priming of T cells. In basal-like breast cancer, nuclear β -catenin (from WNT activation) is linked to low TIL counts. If a tumor upregulates Wnt/ β -catenin (via mutations or autocrine Wnt ligands), it can resist immunotherapy by preventing the initial immune response. *Rationale:* Upregulated Wnt signaling in tumors leads to secretion of factors that repel immune cells or keep them dysfunctional (e.g. via **CCL2** and macrophage polarization), enabling the tumor to escape despite PD-1 being blocked ²¹.
- **PI3K/AKT and MAPK Pathways:** Oncogenic signaling pathways like **PI3K-AKT** and **RAS/MAPK**, when upregulated, contribute to an immune-resistant phenotype. PTEN loss (activating PI3K) has been associated with resistance to PD-1 inhibitors in metastatic ER+/HER2- breast cancer and other cancers ²¹. Upregulated PI3K signaling leads to increased production of immunosuppressive cytokines (e.g. VEGF, IL-8) and fewer TILs. Similarly, hyperactive MAPK pathway (e.g. through KRAS mutations, though rare in breast) can drive expression of PD-L1 but in a context that is non-productive (or cause exclusion of T cells via VEGF). *Rationale:* These pathways sustain tumor proliferation and survival even in the face of immune attack, and they modulate the microenvironment to be less favorable for immune cells (for example, AKT activation in tumor cells upregulates PDL1 and downregulates antigen presentation in some models ²⁹ ²¹). Thus, their upregulation is a hallmark of primary or acquired resistance.
- **Immunosuppressive Cells & Cytokines:** Upregulation of pathways that recruit or activate immunosuppressive cells can mediate resistance. For instance, tumors that highly express **CCL2/CCR2 or CSF1** will attract macrophages (TAMs) that can inhibit T-cell function. Similarly, upregulation of **IL-10** and **IDO** (often downstream of TGF- β and interferon signaling in a feedback loop) creates a local environment that shuts down effector T cells. Breast tumors progressing on PD-1 inhibitors sometimes show increased infiltration of FOXP3⁺ regulatory T cells and CD68⁺ macrophages, implicating the upregulation of chemokine pathways like CCL22/CCR4 (Treg recruitment) in resistance ²⁸. *Rationale:* By turning on these pathways, the tumor essentially seeds its microenvironment with cells and factors that negate the reinvigorated T cells, allowing cancer growth to resume.

Resistance Pathways – Downregulation

Pathways whose *downregulation* (loss of activity) can confer resistance to dostarlimab, usually by disabling critical immune processes:

- **Antigen Presentation (MHC I) Loss:** One of the clearest resistance mechanisms is tumor **downregulation of MHC class I** antigen presentation. Mutations in β_2 -microglobulin (B2M) or epigenetic silencing of antigen-processing machinery (TAP, tapasin) result in tumor cells that no longer display neoantigens to T cells. There are documented cases in melanoma and likely in breast cancer where patients who initially responded to PD-1 therapy relapsed with B2M-mutated, MHC I-null tumors. In breast cancer, an analysis noted that non-responding metastatic lesions had significantly lower HLA-A expression than pre-treatment tumors ¹⁹. *Rationale:* Without MHC I, CD8⁺ T cells (the main effectors unleashed by PD-1 blockade) cannot recognize or kill tumor cells, rendering dostarlimab ineffective. Thus, loss or downregulation of the antigen presentation pathway is a potent escape route.

- **JAK/STAT Signaling Defects:** Tumors that acquire mutations in **JAK1** or **JAK2** can become unresponsive to interferon signals, thereby resisting PD-1 therapy. While this has been reported in other cancers (e.g. melanoma), the principle applies to breast cancer: a JAK1/2 loss-of-function means IFN- γ released by T cells cannot induce PD-L1 or MHC I on tumor cells. Such tumors may not be “seen” by T cells or may not undergo growth arrest in response to immune cytokines. A case report of metastatic TNBC resistant to pembrolizumab revealed a JAK2-inactivating mutation as a possible cause (paralleling known mechanisms). *Rationale: Downregulation of IFN signaling* (due to JAK pathway inactivation) allows the tumor to ignore the immune attack – no upregulation of antigen presentation or apoptosis in response to T-cell cytokines, hence continued survival.
- **Costimulatory Molecule Downregulation:** Tumors or APCs might downregulate costimulatory ligands (like B7-1/B7-2, 4-1BBL, OX40L) as a subtle means of immune evasion. If dendritic cells in the tumor microenvironment lack these signals, T cells remain anergic even if PD-1 is blocked. While less documented in clinical samples, this concept is part of the broader immune escape repertoire: by *not providing sufficient costimulation*, the tumor environment can render T cells ineffective. Some breast cancers have low CD80/CD86 expression on infiltrating antigen-presenting cells – this could be exacerbated after therapy if, say, PD-1 blockade leads to DC exhaustion or clearance. *Rationale:* PD-1 blockade cannot fully activate T cells that never received proper priming; thus a loss of costimulatory pathways equates to primary resistance.
- **Loss of Tumor IFN Receptors or Pathway Members:** Aside from JAK mutations, tumors might downregulate IFN- γ receptor or downstream transcription factors (like IRF1) needed for executing the effects of T-cell cytokines. Reduced expression of IFNGR1/2 has been noted in some resistant contexts. Similarly, downregulation of **IRF1** (a key driver of PD-L1 and CXCL9 expression in response to IFN) can blunt the cancer cell's immunogenic response. Without these pathways, even an active T cell in the vicinity may fail to induce the “death signals” in the tumor. *Rationale:* It's another way tumors make themselves invisible or impervious to immune attack – by silencing the pathway that translates immune signals into tumor cell growth arrest or vulnerability.
- **Apoptotic Pathway Defects:** If a tumor cell downregulates pro-apoptotic mediators or overexpresses anti-apoptotic factors, it might resist T cell killing. For example, loss of **Fas receptor (FAS)** or downregulation of death receptors on tumor cells can lead to resistance, as T cells often induce apoptosis through the Fas/FasL pathway. Some breast cancer subclones under immune pressure might shed or mutate Fas, avoiding this route of cell death. Similarly, high BCL-2 or MCL-1 (anti-apoptotic proteins) in tumor cells can make them less susceptible to perforin/granzyme-triggered apoptosis. *Rationale:* Even if T cells are present and attacking, a tumor that cannot undergo apoptosis readily (due to these pathway alterations) will be relatively resistant to immunotherapy-induced cell death.

(Overall, resistance to dostarlimab in breast cancer is multifactorial. Tumors can avoid immune recognition (antigen presentation loss), prevent immune cell infiltration or function (TGF- β , Wnt, PI3K upregulation), or adapt to neutralize the reactivated T cells (alternate checkpoints up, survival pathway upregulation). These resistance mechanisms underscore the need for combination treatments addressing each pathway: e.g., adding TIGIT/LAG3 inhibitors for checkpoint upregulation, or using epigenetic drugs to restore antigen presentation in MHC-deficient tumors. Recognizing these pathways in a patient's tumor (through biomarkers) is crucial for tailoring therapy after dostarlimab resistance.)

Breast Cancer Subtype–Stratified Evidence

Breast cancer is heterogeneous, and dostarlimab's effects can vary by subtype. Key findings by subtype include:

- **Triple-Negative Breast Cancer (TNBC):** TNBC is the subtype with the most immunotherapy data. It's typically high in PD-L1 expression and TILs. Clinical trials (e.g. KEYNOTE-355, IMpassion130) showed that adding PD-1/L1 inhibitors to chemotherapy improves outcomes *primarily in PD-L1-positive TNBC* ⁴. Dostarlimab is not yet standard in TNBC, but as a PD-1 class drug it's expected to behave like pembrolizumab. Evidence: a meta-analysis of PD-(L)1 trials in advanced TNBC confirmed improved survival, especially in PD-L1⁺ patients ⁴. In early TNBC, neoadjuvant PD-1 inhibition plus chemo significantly raises pCR rates (e.g. ~60% vs 40% in KEYNOTE-522). Biomarker analyses show that **basal-like** TNBC responds better than mesenchymal subtypes, and high TIL counts predict benefit. On the other hand, TNBC tumors with high TGF- β or lacking immune infiltration derive minimal benefit ²². Ongoing trials (IMpassion132, KEYNOTE-552) and real-world studies are evaluating dostarlimab in TNBC, particularly in those who progress on first-line pembrolizumab or in PD-L1⁺-cases combined with other immunomodulators.
- **HR+/HER2- (Luminal A & B):** These estrogen receptor-positive tumors are generally “colder” immunologically. Single-agent checkpoint inhibitors have low response rates (<10%) in metastatic HR+ breast cancer. However, select subgroups show promise: e.g., HR+ tumors with **gBRCA1/2 mutations** or high TMB might respond due to higher neoantigen loads. A recent Phase II trial (TBCRC 056 arm C) in **ER+/HER2-, BRCA-mutated** patients gave neoadjuvant niraparib + dostarlimab; it achieved a 17% pCR rate and 44% near-pCR ⁷ ³⁰ – notable for this subtype. This suggests immunotherapy can work in luminal cancers if DNA repair is compromised (HRD) and an immune response can be stimulated ⁸. Additionally, an exploratory study reported that adding pembrolizumab to chemo in early high-risk Luminal B cancers significantly improved pCR from ~13% to 27% ⁶, indicating some chemo-immune synergy even in ER+ disease. Biomarker insights: luminal tumors with high immune gene expression (an “immune hot” subset ~5–10% of cases) had better than expected responses to ICIs in trials. Conversely, most luminal cancers have active estrogen signaling which can suppress immune responses (e.g. via IL-6/STAT3); ongoing studies are examining if therapies like CDK4/6 inhibitors (which have immunomodulatory effects) combined with PD-1 blockade can enhance efficacy in ER+ disease.
- **HER2-Positive:** HER2⁺ breast cancers are moderately immunogenic; they often have substantial TILs, and some anti-HER2 therapies can induce immune responses (ADCC). Combining PD-1 inhibitors with HER2-targeted therapy is a logical strategy. The Phase III KEYNOTE-811 trial in HER2+ *gastric* cancer (trastuzumab + chemo $\pm$ pembrolizumab) showed improved response rates ³¹, which has influenced breast cancer research. In breast cancer, early-phase data: in the PANACEA study (phase Ib/II), pembrolizumab plus trastuzumab had a 15% response rate in PD-L1⁺ metastatic HER2+ patients who had progressed on trastuzumab ¹⁷. Another trial (KATE2) combining atezolizumab with T-DM1 in advanced HER2+ was negative overall, but hinted that PD-L1⁺ patients might benefit. Neoadjuvant trials are ongoing: e.g., I-SPY2 showed that adding pembrolizumab to chemo + HER2-targeted therapy improved pCR in HER2+ (especially in hormone receptor-negative, HER2+ cases). Biomarker notes: HER2-enriched intrinsic subtype (which is often ER-/HER2+) has high immune activation and may derive the most benefit, whereas luminal HER2+ (ER+/HER2+) tumors are less immune-infiltrated. Also, PI3K pathway mutations (common in HER2+ disease) might limit

immunotherapy efficacy by immune-evasion mechanisms ²¹ , so combinations with PI3K inhibitors are of interest.

- **BRCA1/2-Mutated and HRD-Positive:** This spans across subtypes (many BRCA1 are TNBC, BRCA2 often luminal B). **BRCA-mutant breast cancers** have DNA repair defects leading to genomic instability and potential immune visibility. They also accumulate lymphocytes; for instance, BRCA1-associated TNBCs usually have higher TILs than sporadic TNBCs. Clinical evidence: the aforementioned neoadjuvant PARP + dostarlimab trial in BRCA-mutant ER+ patients showed immunotherapy activity in a setting where normally it would be minimal ⁷ ⁸ . Another example: a small cohort of metastatic BRCA1/2-mutant breast cancer patients had improved disease control with checkpoint inhibitors in some reports (e.g., case series using pembrolizumab post-platinum). It's hypothesized that HRD tumors may respond better due to more neoantigens and cGAS-STING activation from DNA damage ⁸ . However, not all HRD tumors respond – some BRCA2 luminal tumors remain immune-cold. Ongoing trials like KEYNOTE-BRCA (PD-1 ± PARP inhibitor in BRCA-mutated metastatic BC) will shed light. Additionally, **PALB2-mutated** breast cancers (which are HRD) were included with BRCA in TBCRC 056 and showed similar immune changes ⁷ .
- **Basal-like vs. Other TNBC Subtypes:** Within TNBC, intrinsic subtypes matter. **Basal-like** TNBC (which overlaps with immunomodulatory subtype) tends to respond to ICIs if PD-L1⁺. In contrast, **mesenchymal** and **luminal-androgen receptor (LAR)** subtypes of TNBC have lower response rates. Basal-like tumors often have high TILs and PD-L1; in the IMpassion130 trial, patients with basal-like signatures had the largest benefit from atezolizumab. **PD-L1 expression** is highest in the basal-like/immune-activated subgroup. LAR TNBCs (which express AR) are usually cold; interestingly, a clinical trial of AR inhibitor + pembrolizumab is exploring if turning off AR can inflame these tumors. **Claudin-low** breast cancers (which are often TNBC or metaplastic) have high immune infiltration but also high immunosuppressive features (Tregs, checkpoint expression); some have responded to ICIs, but others progress rapidly (likely due to macrophage/TGF-β-rich milieu). For metaplastic TNBC, small studies reported relatively high response rates (~20–30%) to PD-1 inhibitors, suggesting this rare subtype (often PD-L1⁺) may be sensitive.
- **Luminal A vs. Luminal B:** Among HR+ cancers, **Luminal B** (high Ki-67, more HER2-enriched gene expression) has slightly more immune infiltration than Luminal A. It's hypothesized that Luminal B might gain a bit more from immunotherapy, especially in the high-proliferation, endocrine-resistant setting. The I-SPY2 trial arm with pembrolizumab did show some pCR improvements in ER+ (most of whom were Luminal B) ⁶ . Luminal A, being lower grade and low proliferation, rarely has substantial TILs and has shown almost no ICI activity in metastatic trials. In early-stage, if Luminal A tumors are low-risk, immunotherapy is not being considered. In metastatic luminal disease, combinations like pembrolizumab with endocrine therapy did not show clear benefit (KEYNOTE-028 and JAVELIN trials had minimal responses in ER+ cohorts). Research is ongoing into whether induced hypermutation (e.g. using APOBEC induction or CDK4/6 inhibitor's immunomodulatory effect) could render Luminal A tumors more immunogenic.
- **HER2-Enriched vs. Other HER2+:** "HER2-enriched" is an intrinsic subtype (by gene expression) found in some HER2+ tumors (and some HER2-). HER2-enriched tumors have higher immune signatures on average than the Luminal HER2 subtype. Post-hoc analyses from the KATE2 trial hinted that patients with HER2-enriched profiles had numerically better outcomes with atezolizumab + T-DM1 compared to T-DM1 alone, whereas those with luminal profiles did not – aligning with the idea that the HER2-

enriched subtype (often non-luminal) is more immunotherapy-responsive. Additionally, **ERBB2-driven** oncogenic pathways can induce PI3K/AKT; patients whose HER2+ tumors also carry PIK3CA mutations might respond less to immunotherapy due to PI3K-mediated immunosuppression ²¹. There's interest in combining PI3Kα inhibitors with ICIs in this subset.

In summary, **PD-1 blockade is most established in TNBC**, conditionally useful in HER2+, and investigational in HR+ breast cancer. Within each category, biomarkers like PD-L1 status, TIL levels, BRCA/HRD status, gene expression subtype, and oncogenic co-mutations refine who is likely to benefit. Ongoing clinical trials are further defining these subtype-specific roles of dostarlimab and similar agents.

Contraindications and Safety

Contraindications: According to the FDA label, Jemperli (dostarlimab-gxly) has **no absolute contraindications** except for severe hypersensitivity to the drug or its components ³². The EMA SmPC likewise lists only hypersensitivity as a contraindication ³³ ³⁴. In practice, this means no patient subgroups are outright excluded from dostarlimab solely due to comorbid conditions, aside from a history of anaphylaxis to similar monoclonal antibodies. There are **no contraindications related to hepatic or renal impairment** explicitly; however, these populations were not extensively studied (use with caution). Pregnancy is not a formal “contraindication” in labeling, but **use in pregnancy is strongly advised against** due to risk of fetal harm from immune activation – PD-1 inhibitors can cause immune-mediated rejection of the fetus and are teratogenic in animal models ³⁵. Breastfeeding is similarly not recommended during and for months after therapy. Patients with active autoimmune diseases requiring immunosuppression were excluded in trials; while not a label contraindication, initiating dostarlimab in such patients is generally avoided or done with extreme caution (risk of exacerbating autoimmune disease).

Safety Profile: Dostarlimab's safety in breast cancer is expected to mirror that observed in other tumors. The chief concerns are **immune-related adverse events (irAEs)** resulting from general immune activation ³⁶ ³⁷. These can affect any organ system: - **Dermatologic:** rash and pruritus (common, usually mild), vitiligo (in TNBC patients, as seen with other ICIs). - **Endocrine:** immune-mediated thyroid disorders are frequent (hypothyroidism in ~10–15% ³⁸), managed with hormone replacement. Less commonly, hypophysitis (pituitary inflammation) can occur, leading to adrenal insufficiency – this is more frequent with CTLA-4 agents but seen with PD-1 inhibitors too. - **Hepatic:** immune-mediated hepatitis (transaminase elevations) occurs in a minority; Grade 3–4 hepatitis in ~1–2%. Monitoring LFTs is required ³⁹. - **GI:** immune-mediated colitis is a serious toxicity (incidence ~1–2% for Grade ≥3). Diarrhea is relatively common (any grade ~10%); severe colitis is rarer but can be life-threatening, often requiring steroids. - **Pulmonary:** pneumonitis has been observed (in <5% of patients; Grade 3–4 ~1–2%). Prior thoracic radiation may predispose to pneumonitis ⁴⁰. Any new cough or dyspnea on therapy warrants evaluation, as immune pneumonitis can be fatal if not treated promptly ³⁶. - **Others:** immune-mediated nephritis, pancreatitis, uveitis, myocarditis are rare (<1% each) but reported in class. Infusion reactions can occur (usually mild chills/fever; severe anaphylaxis is rare).

In the context of breast cancer combination therapies (e.g. with chemo or PARP inhibitors), overlapping toxicities need attention. The combination of dostarlimab with chemotherapy can increase risks of neutropenia and pneumonitis (some chemo agents cause lung toxicity, e.g. gemcitabine, which combined with an ICI might elevate pneumonitis risk). Combining with PARP inhibitors (as in the gBRCA trial) led to expected PARP-side effects plus immune effects: in TBCRC 056, common Grade ≥3 AEs were rash (28%), liver enzyme elevation (~17%), and nausea (6%) ⁴¹. These were manageable with supportive care and/or

steroids for rash. Notably, **immune-related rash** (maculopapular) was one of the most frequent serious AEs in that trial ⁴¹, consistent with known ICI dermatologic toxicity.

Special Populations: - Patients with prior allogeneic transplants or active autoimmune disorders face higher risk of severe irAEs. While not absolutely contraindicated, such patients should only receive dostarlimab if potential benefit far outweighs risk, and with close monitoring. For example, a patient with lupus or rheumatoid arthritis might experience flare-ups or even life-threatening immune events on PD-1 therapy. - Organ transplant recipients on immunosuppressives would likely reject the graft if given dostarlimab (due to immune activation), so this is generally avoided. - There is limited data in patients with HIV or chronic viral infections: these were excluded from trials. Current practice is to allow well-controlled HIV patients on therapy with close observation (no contraindication per se, but ensure CD4 counts are adequate). - Geriatric use: No significant differences in safety or efficacy were seen in older patients (65+), per available data ⁴². However, older patients may have less robust immune responses and possibly lower irAE incidence, but also may tolerate severe irAEs less well. - No dose adjustments are recommended for mild-to-moderate renal or hepatic impairment ⁴³ ⁴⁴, but data in severe impairment are lacking.

In summary, **safety management** revolves around early identification of irAEs ⁴⁵. Guidelines call for regular monitoring of thyroid function, liver enzymes, and clinical symptoms. Corticosteroids are the mainstay for managing moderate-to-severe irAEs, and treatment interruption or cessation is indicated for Grade 3–4 toxicities ⁴⁶. Importantly, per FDA label, **dostarlimab should be permanently discontinued for any life-threatening irAE** (e.g. Grade 4 pneumonitis, neurologic toxicity) and withheld for moderate-to-severe events until improvement ⁴⁷. There are **no black box warnings** for dostarlimab; however, clinicians are alerted to immune-mediated reactions in the warnings/precautions section ³⁶.

Trial and Guideline Context

Clinical Trials: Although dostarlimab itself is newer in breast cancer trials, multiple studies explore PD-1/PD-L1 blockade in this setting: - *Metastatic TNBC*: KEYNOTE-355 (pembrolizumab+chemo) and IMpassion130 (atezolizumab+nab-paclitaxel) established the benefit in first-line PD-L1⁺ TNBC ⁴. By analogy, dostarlimab is being tested in TNBC as well. For instance, GSK has initiated trials combining dostarlimab with chemotherapy in PD-L1⁺ metastatic TNBC to seek similar indications. Additionally, the I-O combinations like NCT04109066 evaluate dostarlimab with novel agents in refractory TNBC. - *Neoadjuvant/Early TNBC*: Trials like I-SPY2 included pembrolizumab; results showed a significant pCR boost. Dostarlimab is part of ongoing neoadjuvant platforms (e.g., GSK is collaborating in the I-SPY2 trial with an arm of dostarlimab + oral paclitaxel ± trastuzumab in early TNBC and HER2+; preliminary results awaited ⁴⁸). Also, the NEOTORCH trial (NCT04358922) is evaluating toripalimab (a PD-1 antibody) in early TNBC – relevant for class effect. - *HR+/HER2– advanced*: The phase III KEYNOTE-756 (pembrolizumab + endocrine therapy ± chemo in high-risk early HR+ BC) results are pending; interim analysis did show higher pCR with pembro in a subset ⁶. For metastatic HR+ BC, trials like JAVELIN Medley (avelumab + axitinib in ER+), and KATE3 (atezo + hormonals) have been largely negative or inconclusive. One interesting trial is NCT04584255 (TBCRC 056), which tested **niraparib + dostarlimab preoperatively in BRCA/PALB2-mutant breast cancer**: Arm C (ER+/HER2–) showed evidence of activity (pCR 16.7%) ⁴⁹ ⁵⁰ and immune biomarker modulation (TIL increase) ⁵¹; Arms A/B (TNBC with differing combinations) are maturing. This TBCRC trial suggests non-chemotherapy immunotherapy strategies can work in biomarker-selected patients and informs future designs. - *HER2+ disease*: KEYNOTE-811 in gastric is a proof-of-concept; in breast, the analogous trial is ongoing (pembrolizumab + trastuzumab + chemo in first-line HER2+ metastatic; interim safety is acceptable). The PANTHER trial (NCT02914470) is testing nivolumab + trastuzumab in HER2+ breast

cancer. Also, neoadjuvant trials like I-SPY2 and NSABP B-59 (atezo + chemo in early TNBC/HER2-) indirectly support IO in HER2+ by showing it's safe to combine with common chemo backbones (many HER2+ regimens include the same taxanes). - *Combination trials*: Many are exploring synergy: e.g., PARP inhibitors (MEDIOLA trial had durvalumab + olaparib in BRCA-mut BC), anti-LAG3 or TIGIT with PD-1 in TNBC (NCT04162127, etc.), MEK inhibitors plus PD-1 in TNBC (to modulate tumor microenvironment). A trial of **dostarlimab + carboplatin/paclitaxel ± bevacizumab** in PD-L1⁺ TNBC (phase II) is recruiting to see if combining anti-VEGF yields further benefit. - *Tissue-agnostic use*: Dostarlimab is approved for dMMR solid tumors ²⁵. Though primary breast cancer rarely is dMMR (~1–2%), metastatic lesions can acquire MSI-H status. There have been case reports of dramatic responses to PD-1 inhibitors in MSI-H breast cancer. Dostarlimab could be used in this setting as an alternative to pembrolizumab. Similarly, TMB-high (≥10 mut/Mb) tumors are approved for pembrolizumab; presumably dostarlimab would work as well, though it's not label-approved for TMB-high specifically.

Guidelines: As of 2024, NCCN and other guidelines for breast cancer specifically list pembrolizumab (and formerly atezolizumab) for PD-L1⁺ advanced TNBC and pembrolizumab in early-stage TNBC with high risk features. Dostarlimab is not named in NCCN Breast Cancer Guidelines for TNBC, as it isn't FDA-approved in this indication. However: - **NCCN (2023)**: Recommends considering a PD-1 inhibitor in metastatic TNBC if PD-L1 CPS ≥10 (using pembrolizumab as the example) ⁴. For MSI-H metastatic breast cancer (any subtype), NCCN mentions pembrolizumab as a category 2A option. By extension, dostarlimab (with its FDA nod for dMMR tumors) could be considered in dMMR breast cancer if pembrolizumab isn't available or as part of a trial. NCCN also lists clinical trials as preferred for PD-L1-negative TNBC, many of which involve PD-1/L1 agents in novel combinations. - **ESMO**: The ESMO clinical practice guideline (2021) echoes that immunotherapy is standard in first-line PD-L1+ TNBC (with pembro+ chemo). For other subtypes, ESMO currently does not endorse immunotherapy outside of trials, with a note that HR+ patients should not receive ICIs off-trial due to lack of proven benefit. - **ASCO**: ASCO's guideline for metastatic TNBC (2022) recommends offering pembrolizumab + chemo for PD-L1 CPS ≥10 TNBC. It also emphasizes germline BRCA testing (for PARP inhibitors) and suggests that outside of PD-L1+ TNBC or MSI-H tumors, ICIs should be given in trial context. Dostarlimab, being similar to pembro, might be considered for MSI-H cases if pembro is contraindicated or unavailable. - **FDA approvals relevant**: Pembrolizumab is approved for early TNBC (neoadjuvant/adjuvant) and for metastatic PD-L1+ TNBC; atezolizumab previously had accelerated approval for PD-L1+ TNBC (withdrawn in 2021 in the US). Dostarlimab's approvals in endometrial/dMMR are noted in FDA communications ². There is currently no FDA or EMA approval of dostarlimab specifically for breast cancer. However, its use in clinical trials and off-label for MSI-H breast cancer is within rationale due to class effect.

Notable Trial Results: - **TBCRC 056 (2023)**: As detailed, it provided proof that even ER+ BRCA-mutated tumors can respond to a PARP+PD-1 regimen, expanding the scope of immunotherapy in breast cancer ⁴⁹⁵⁰. - **SABCS 2024 Highlights**: The trial above (Mayer et al.) was presented at SABCS 2024, underscoring interest in chemo-free immunotherapy strategies for subsets of breast cancer ⁴⁹. Also at SABCS/ESMO, updates on KEYNOTE-522 (early TNBC) showed pembrolizumab significantly improves event-free survival, solidifying IO in curative TNBC. - **NeoTRIPaPDL1 trial**: Although it tested atezo in early TNBC without improving pCR, an interesting biomarker was that patients with high immune gene expression did well regardless, whereas those with low immune signatures did poorly even with IO. This reinforces patient selection by immune pathways. - **DESTINY-Breast03 Immune analysis** (T-DXd ± nivo): though T-DXd is antibody-drug conjugate, combination with immunotherapy is being explored given ADCs may induce immunogenic cell death.

Emerging directions: Guidelines are likely to evolve. For example, if ongoing trials like KEYNOTE-B-12 (adjuvant pembro in residual TNBC) or KEYNOTE-756 (neoadjuvant pembro in high-risk HR+) are positive, immunotherapy usage will broaden. The role of dostarlimab specifically might grow if head-to-head trials or new approvals occur, or if it's bundled in combo approvals (for instance, a PARP+PD-1 combo indication for BRCA-mutated cancer could potentially involve dostarlimab given GSK's interests). Multidisciplinary tumor boards now discuss checkpoint inhibitors for oligometastatic or heavily pretreated cases with high TILs or MSI-H findings, even beyond label – these discussions often reference pembrolizumab data, but in principle apply to any PD-1 agent.

In conclusion, **dostarlimab in breast cancer** is currently positioned as an investigational immunotherapy being assessed in trials across subtypes, with its use informed by the successes of other PD-1/L1 inhibitors. Guidelines endorse PD-1 blockade in PD-L1+ TNBC and dMMR tumors; for other subtypes, its use is limited to clinical trials or exceptional cases. As research continues (especially in combinations exploiting specific pathways), we anticipate more nuanced recommendations that incorporate subtype, biomarkers, and rational combos (e.g. adding checkpoint blockade in select ER+ cases with BRCA mutations or considering dual checkpoint blockade in “cold” tumors).

Additional Mechanistic/Clinical Notes

- **Biomarker Development:** There is intense research into predictive biomarkers beyond PD-L1 IHC. In breast cancer, gene expression profiles (like the 18-gene T-cell-inflamed GEP ⁵²), TIL quantification, and circulating immune markers (e.g. neutrophil-to-lymphocyte ratio) are being studied. One interesting finding is that **baseline stromal TIL level** is predictive of pCR and survival benefit from checkpoint inhibitors across trials ⁵¹. Higher baseline TILs not only correlate with PD-L1 positivity but also with genomic features (e.g. HRD, TP53 mutations) that may underlie an immune-responsive phenotype ⁵³. Conversely, tumors with “immune deserts” on biopsy are very unlikely to benefit from dostarlimab; trials like FUTURE-C-PLUS stratify patients by immune phenotype to tailor therapy.
- **Immune Microenvironment Modulation:** There is interest in using therapies that modulate the TME to improve ICI efficacy in breast cancer. For example, **radiotherapy** can have an abscopal effect – low-dose radiation to a tumor site can upregulate MHC and chemokines, potentially converting an ICI-nonresponsive tumor to responsive. An NIH trial (NCT03711058) is testing radiation + pembrolizumab in metastatic TNBC. Early results show some abscopal responses, suggesting radiotherapy upregulates pathways (like Type I IFN via cGAS–STING) that sensitize tumors to PD-1 blockade ⁵⁴.
- **Vaccines and Adoptive T Cells:** In HER2+ breast cancer, there are vaccines (like NeuVax) and cellular therapies (HER2 CAR T cells) being studied; combining these with PD-1 blockade might be synergistic. The rationale is that a vaccine or CAR T provides a bolus of tumor-specific T cells or antigen priming, and PD-1 inhibitor sustains their function in vivo. Trials combining a mammaglobin-A DNA vaccine with pembrolizumab in metastatic HR+ breast cancer (NCT03199040) are ongoing.
- **Unique Breast Cancer Immune Features:** Unlike melanoma or lung cancer, most breast cancers (especially ER+) have relatively low mutation load. However, they often have **oncogenic virus-like transposon signatures (e.g. endogenous retroviruses)** that can trigger innate immunity when unmasked (a phenomenon exploited by epigenetic drugs). Some studies use DNA methyltransferase inhibitors in ER+ breast cancer to induce viral mimicry, thereby recruiting immune cells and then add PD-1 blockade.

- **Safety in Combination:** It's worth noting that combining PD-1 inhibitors with other therapies raises new safety considerations. For instance, combining with PARP inhibitors (as with niraparib + dostarlimab) increases risk of anemia, liver toxicity and potentially pneumonitis (as both can rarely cause pneumonitis). Combining with anti-HER2 agents can raise concerns about cardiac toxicity (both trastuzumab and pembrolizumab can cause myocarditis rarely). No alarming synergistic toxicity has emerged so far, but vigilance is needed. In the TBCRC 056 trial, the combination was overall tolerable but did have ~28% Grade 3 rash ⁴¹ (perhaps more than either alone, indicating an interaction in immune side effects).
- **Cross-Talk with Hormone Signaling:** There's an evolving understanding that estrogen signaling may modulate anti-tumor immunity. ER activation in breast cancer cells can upregulate PD-L1 via PI3K signaling and also drive secretion of T cell suppressive cytokines (like TGF- β , IL-4). Conversely, endocrine therapy (aromatase inhibitors, SERDs) might slightly enhance T-cell activity by removing those signals. Some preclinical data suggest fulvestrant (an ER degrader) plus anti-PD-L1 slowed tumors more than either alone in ER+ models. These interactions are being explored in trials like NCT02997995 (atezo + fulvestrant).
- **Immune-Related Pathologic Response (irPR):** Pathologists evaluating resected specimens after neoadjuvant immunotherapy note patterns like "immune cell-only residuals" (tumor replaced by immune granulomas or fibrosis). This may call for new pathological response criteria. In TNBC, residual cancer burden (RCB) is used; interestingly, some patients with residual tumor post-ICI still do very well if that tumor is heavily infiltrated by activated T cells (perhaps indicating an ongoing immune kill that might clear micrometastases). The phenomenon of pCR in mismatch-repair deficient cancers with dostarlimab (as seen in rectal cancer ⁵⁵) raises the question: could a subset of breast cancers (like MMR-deficient) achieve complete response without surgery? Currently, surgery remains standard in breast cancer, but these dramatic results in other tumors spur hypothesis-generating discussion.
- **Economic and Access Considerations:** While not mechanistic, it's notable that checkpoint inhibitors, including dostarlimab, are expensive. In health technology assessments (like CADTH in Canada ¹⁴), cost-effectiveness in breast cancer is scrutinized. As dostarlimab seeks new indications, demonstrating clear biomarker-enriched efficacy will be key to justify its use. For now, payers typically cover pembrolizumab for eligible breast cancer cases; dostarlimab might be substituted in certain situations (e.g., patient in trial, or if a patient developed an infusion reaction to one PD-1 antibody, sometimes another can be tried).
- **Future Directions:** Combinations are the future – e.g., **triplet therapy** like chemotherapy + PARP inhibitor + PD-1 (there is a trial with nivolumab + niraparib + cisplatin in TNBC). Also, **bispecific antibodies** (e.g., one arm anti-PD-1, another anti-TGF- β or anti-LAG3) are in development – these could potentially supplant the need for multiple drugs. For instance, **M7824** (bintrafusp alfa) is a bifunctional anti-PD-L1/TGF- β trap that was tested in TNBC. While it didn't yet beat standard therapy, such approaches aim to tackle multiple resistance pathways at once. GSK itself is testing cobolimab (an anti-TIM3, formerly TSR-022) with dostarlimab in various tumors ⁵⁶; should TIM-3 prove a major resistance factor in breast cancer, that combo could be a logical next step.
- **Real-World Evidence:** Early real-world data from expanded access programs suggest that some heavily pretreated metastatic breast cancer patients (particularly TNBC) had prolonged stabilization with PD-1 inhibitors beyond trial settings, though response rates are low. Ongoing registries will provide insight into how patients not fitting trial criteria (e.g. older, comorbid) fare on immunotherapy. Additionally, diversity in clinical trials is an issue: most immunotherapy trials had underrepresentation of certain racial groups that have higher TNBC incidence. Efforts to broaden participation are important to confirm that these findings generalize.

In summary, dostarlimab's incorporation into breast cancer care will rely on identifying the right patients (likely via multi-parameter biomarkers) and the right partners (combination therapies) to maximize immunogenic potential. The landscape is rapidly evolving, and lessons from across oncology are being applied to transform breast cancer – historically not the most immune-responsive tumor – into one where immunotherapy can play a significant role in improving outcomes.

Final Pathway Evidence Table

Pathway (ID/Name)	Regulation	Effect of Dostarlimab	Biological Rationale	Referen
PD-1 Checkpoint Signaling (<i>REACTOME_CO_INHIBITION_BY_PD_1</i>)	– (Blocked)	Sensitive (MoA)	Blocking PD-1 lifts T-cell inhibition, restoring anti-tumor activity. PD-1/PD-L1 pathway inhibition leads to T-cell activation and tumor cell killing in responsive breast tumors. ³	³
T-cell Activation (TCR pathway) (<i>KEGG hsa04660</i>)	Up	Sensitive	Robust TCR signaling and antigen presentation enhance ICI efficacy. Tumors with intact/active T-cell receptor pathways (high MHC-I, costimulation) respond better to PD-1 blockade. ¹⁹	¹⁹

Pathway (ID/Name)	Regulation	Effect of Dostarlimab	Biological Rationale	Reference
Interferon-γ Response (<i>HALLMARK_INTERFERON_GAMMA_RESPONSE</i>)	Up	Sensitive	Th1 cytokine milieu promotes tumor recognition. IFN- γ -driven gene signature (CXCL9/10, MHC upregulation) correlates with PD-1 inhibitor responses in TNBC (immune-“hot” tumors). ⁴	⁴
cGAS–STING Pathway (<i>KEGG hsa04623</i>)	Up	Sensitive	Cytosolic DNA sensing boosts Type I IFNs and dendritic cell activation. STING upregulation (e.g. via DNA damage in HRD tumors) recruits T-cells, improving dostarlimab efficacy. ⁸	⁸
Antigen Presentation (MHC I) (<i>GO:0019882</i>)	Down (loss)	Resistant	Loss of MHC-I prevents T-cell recognition of tumor. Tumors escaping ICI often have downregulated antigen-presenting machinery (e.g. B2M mutation), abrogating cytotoxic T-cell killing. ¹⁹	¹⁹

Pathway (ID/Name)	Regulation	Effect of Dostarlimab	Biological Rationale	Referen
TGF-β Signaling (HALLMARK_TGF_BETA_SIGNALING)	Up	Resistant	TGF- β creates an immune-excluded TME by fibrosis and T_reg/MDSC recruitment. High TGF- β activity (in stroma and tumor) is linked to lack of response to PD-1/L1 blockade in breast cancer. ²²	²²
Wnt/β-Catenin Signaling (KEGG hsa04310)	Up	Resistant	Wnt activation inhibits intratumoral dendritic cells, leading to poor TIL priming. β -Catenin-high basal tumors exclude T-cells and resist checkpoint immunotherapy. ²¹	²¹
PI3K-AKT Pathway (HALLMARK_PI3K_AKT_MTOR_SIGNALING)	Up	Resistant	Oncogenic PI3K signaling drives production of immunosuppressive factors (e.g. VEGF) and reduces TILs. PTEN loss ($\uparrow$ PI3K) in breast tumors correlates with diminished ICI efficacy ²¹	²¹

Pathway (ID/Name)	Regulation	Effect of Dostarlimab	Biological Rationale	Referen
TIM-3 Checkpoint Pathway (<i>HAVCR2</i> upregulation)	Up	Resistant	Compensatory upregulation of TIM-3 on T-cells leads to continued T-cell exhaustion despite PD-1 blockade. PD-1 therapy can induce TIM-3 via PI3K/Akt, allowing tumor immune escape. ²³	²³
IDO1/Kynurenine Pathway (<i>Tryptophan</i> catabolism)	Up	Resistant	IDO depletes tryptophan and generates immunosuppressive metabolites, disabling effector T-cells. Tumors with high IDO1 (often driven by IFN- γ) create a local tolerance, blunting PD-1 inhibitor effect. ²⁸	²⁸
IL-10/Immunosuppressive Cytokines (<i>Immune suppressive milieu</i>)	Up	Resistant	Elevated IL-10, TGF- β and other suppressive cytokines maintain a “cold” TME. Breast tumors with an immunosuppressive cytokine profile (high IL-10, etc.) have limited T-cell activation under ICI. ⁵⁷ ²²	⁵⁷ ²²

Pathway (ID/Name)	Regulation	Effect of Dostarlimab	Biological Rationale	Referen
JAK-STAT/IFN Signaling (IFN receptor pathway)	Down (defective)	Resistant	Mutations or downregulation in JAK1/2 or IFN γ R cause tumors to ignore IFN- γ from T-cells. Lack of response to IFN means no upregulation of antigen presentation or apoptosis, conferring resistance (analogous cases seen with JAK mutations in ICI-refractory cancers)	19 *

* 19 – Indicates tumor strategies of immune escape including lowered antigen presentation and checkpoint upregulation (e.g. mechanisms akin to JAK/STAT pathway disruptions leading to immune escape).

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